Night and Day: Diurnal Warming and Structural Transformation in India

Vedarshi Shastry
University of California, Santa Cruz

June 26, 2026

Abstract

This paper finds diverging partial effects of diurnal warming – higher nighttime (T_{min}) and daytime (T_{max}) temperatures – on the share of agricultural wage-labor in decadal Indian Censuses (1981–2011). Both warmer nights and days contract cultivated area and grain output; only warmer days raise harvest prices locally. Warmer nights consolidate agriculture, shifting seasonal workers and self-cultivators into wage labor; warmer days casualize it, pushing wage labor back to seasonal work. Long differences show the labor divergence is significant only in rural populations. In towns, both margins depress the non-agricultural share.

Keywords: Climate Change, Structural Transformation, Agriculture, India
JEL Codes: O1, Q54, J43, R11

Working paper [arXiv]. This research was conducted and written independently with no conflict of interest. All errors are my own. I am grateful to Aaditya Dar and seminar participants at the World Bank and UC Santa Cruz for their feedback over various stages of the project. This paper has also benefited from agentic research assistance provided by Gemini Pro (2-3.1) and Claude Opus (3-4.8) models.
Email: veshastr@ucsc.edu

The reallocation of labor from agriculture into non-agricultural work—structural transformation—is the foundation of economic development (Matsuyama 1992; Herrendorf, Rogerson, and Valentinyi 2014). Whether a warming climate arrests it is unsettled, and the evidence splits by horizon. In the short run, heat pushes workers off the field: Colmer (2021) finds that year-to-year heat shocks move Indian labor out of agriculture within the year, and in rural Mexico Jessoe, Manning, and Taylor (2017) finds that extreme heat depresses farm and local non-farm employment alike, pushing workers to migrate out. Over the long run the sign flips—reading six decades of the Census, Liu, Shamdasani, and Taraz (2023) finds that sustained warming in India holds labor on the land and depresses non-farm demand, while across countries Nath (2025) shows that when heat lowers farm productivity a "food problem" locks economies into agricultural specialization. Within India, then, the two horizons disagree on the central margin of development.

Yet the escape valve that clears the shock in Mexico is largely shut in India: rural–urban mobility is strikingly low, as households forgo large urban wage gains to keep the caste-based insurance networks that out-migration would sever (Munshi and Rosenzweig 2016). Displaced labor cannot simply leave; it must be absorbed locally. And the stakes are first-order—agriculture still employs close to half of India's workforce, so the direction in which heat moves that labor decides whether warming hastens India's structural transformation or stalls it.

Prior literature measures heat with a daily average, a count of degree days, or a tally of extreme days (Schlenker, Hanemann, and C. 2006; Blakeslee and Fishman 2015; Dell, Jones, and Olken 2014), which sum two partial effects that need not share a sign. The nighttime minimum and daytime maximum have themselves warmed at different rates in India: since the early 1990s the former has risen faster than the latter, narrowing the diurnal range across several agro-climatic zones, including the Indo-Gangetic crop belt (Mall et al. 2021). This compression is geographically organized (Figure A2): the diurnal range is narrowing across the northern Indo-Gangetic plains and the arid northwest, where nights have warmed faster, but widening across the peninsular Deccan and the south, where days have warmed faster—a north/south split in the *direction* of diurnal change that the analysis later exploits. Crops stand in the field around the clock; the farmers that tend to them work only in daylight. A warmer night raises nocturnal respiration and the overnight heat load the crop carries, spending after dark the carbon it fixes by day (Peng et al. 2004; Welch et al. 2010); a warmer day presses field labor against the physiological ceiling of sustained outdoor exertion, lowering the productivity of labor itself (Dunne, Stouffer, and John 2013; Somanathan et al. 2021). These are dry-bulb, air-temperature channels, and the responses estimated here are decadal adaptations rather than single-season yield shocks: over a decade cultivation adjusts to the warming it faces, so the estimates recover the net direction of labor reallocation, not the sign of any one season's yield. This paper holds the two margins apart, estimating the partial effect of the nighttime minimum (T_{min}) and the daytime maximum (T_{max}), each net of the other and of rainfall, across Indian Census districts, towns, and urban households from 1981 to 2011. A higher minimum compresses the diurnal range and a higher maximum widens it; where the daily average collapses the cycle to a single mean, the decomposition recovers its range—the quantity agronomy ties to grain filling and heat stress, and, as the results show, the margin that reallocates labor where the daily level only contracts scale.

I establish three central empirical facts, then two extensions that test how far they reach; the general-equilibrium framework of Section 1 is built on the central facts, its assumptions dictated by the data rather than imposed on it.

First, the composition of agricultural employment reverses between night and day. In a decadal district panel, a 1°C rise in the nighttime minimum raises the agricultural wage-labor share by 4.4 percentage points, drawing workers from self-cultivation (-2.8) and the seasonal margin (-3.3); an equal rise in the daytime maximum reverses the wage-labor and seasonal signs, cutting the wage-labor share by 2.6 points and returning 3.2 to the seasonal pool. The two coefficients reject equality ($F = 24.9$ for wage labor, 14.7 for seasonal), and the reversal is the diurnal cycle's own: a single growing-season degree-days measure collapses it to one muted coefficient ($+0.036$). It survives a two-decade long difference ($+0.075/-0.062$) and humidity-adjusted wet-bulb temperatures ($+0.086/-0.079$).

Second, the two margins move the farm economy asymmetrically. Both contract cultivated area by indistinguishable amounts (-0.22 and -0.22, $F = 0.01$), while only the daytime maximum raises the local harvest price ($+0.13$) and depresses the field wage (-0.12), the nighttime minimum leaving both flat. Daytime heat acts as a classic supply shock—scarcer output, dearer grain, cheaper labor—while nighttime warming rearranges who works the land at an unchanged price.

Third, the reversal scales with the weight of agriculture in local employment: over the two-decade long difference it is sharp among rural workers and barely registers among urban, the amplitude tracking the agricultural share of the workforce (Figure 2; Table A11).

The reach of the decomposition then shows in two extensions. First, in towns the divergence closes: over a long difference, both margins depress the town non-agricultural share—by 3.8 points under warmer nights and 7.8 under hotter days—so each end of the cycle holds labor out of non-agriculture.

Second, and only suggestively, matched to urban households in three rounds of the National Sample Survey, the displaced labor arrives at the informal, casual end of urban work—petty trade, transport, construction—rather than in manufacturing or salaried employment (Appendix Table A16).

What unites these facts is that nighttime and daytime warming reach the farm through different factors of production. A warmer night acts on land, scaling the productive services of the soil; a warmer day lowers total factor productivity, eroding the capacity for sustained physical work. When labor and land are gross substitutes—their elasticity of substitution above one—the two shocks reach labor demand through different channels: a land-augmenting shock alters the land-to-labor ratio and shifts the demand for agricultural labor directly, independent of its effect on yields, whereas a Hicks-neutral productivity loss leaves that ratio intact and contracts only the scale of production. The two thus move agricultural labor demand in opposite directions—warmer nights raise it, drawing self-cultivators and seasonal workers into agricultural wage labor, and warmer days lower it, returning laborers to the seasonal margin. Under unit elasticity, the Cobb–Douglas case, the two shocks are observationally equivalent and the divergence cannot arise; gross substitutability is the economic content of the result.

What discriminates the mechanism is a restriction the labor signs alone do not impose: only the daytime margin moves the local harvest price ($F = 13.3$ on price equality), the nighttime margin leaving it flat. That asymmetry is the signature of partial market integration—a district whose food price is set largely by trade, so that only a productivity-driven output loss passes into the local price. The amplitude of the reallocation then scales with the agricultural share of local employment and the depth of the seasonal margin—large among rural workers, muted among urban. A general-equilibrium treatment in Appendix A formalizes this, spanning open and closed food markets; it is the over-identifying price restriction, not the labor signs, that places India's districts on the integrated side. This is a different object from the closed-economy "food problem" of Liu, Shamdasani, and Taraz (2023) and Nath (2025), which turns on a national, long-run subsistence price spiral; the framework fixes the direction of each response, and the magnitudes are the data's to supply.

This paper makes four contributions. First, the diurnal decomposition reconciles the conflicting short- and long-run estimates of Colmer (2021) and Liu, Shamdasani, and Taraz (2023): warming sorts agricultural labor rather than pushing it in a single direction, and which way it moves depends on which end of the day warms. Second, a general-equilibrium framework shows the night/day divergence to be a property of factor bias under gross substitutability, and carries an over-identifying restriction—only daytime warming moves the local harvest price—that the data confirm ($F = 13.3$), corroborating partial market integration; this is distinct from the national, long-run "food problem" of Liu, Shamdasani, and Taraz (2023) and Nath (2025) rather than a refutation of it. Third, I construct a consistent-boundary panel of Indian towns from the digitized Census and link it to urban household surveys for suggestive evidence on the informal destinations of displaced agricultural labor that district aggregates leave unresolved. Fourth, the paper qualifies the structural-transformation literature (Matsuyama 1992; Herrendorf, Rogerson, and Valentinyi 2014): under sustained heat, towns foreclose entry into non-agricultural employment—a reallocation of labor that stalls, rather than advances, the structural transition. Section 1 develops the framework; the next section describes the data and the decomposition; Section 3 reports the district, town, and household results; and the last concludes.

1. Conceptual Framework

This section lays out the economic logic that maps each temperature margin to the labor responses the rest of the paper estimates. Appendix A formalizes it and Table A1 collects the predicted signs.

What the two margins measure. The nighttime minimum and the daytime maximum are the floor and the ceiling of the daily temperature cycle. Holding one fixed, a higher minimum compresses the diurnal range and a higher maximum widens it, so the decomposition speaks to the range—the quantity agronomy ties to grain filling and heat stress—as much as to the level. The two margins act on different inputs. Crops occupy the field around the clock, so a warmer night works on the standing crop and the land in cultivation, raising respiration and the overnight heat load the crop carries (Peng et al. 2004; Welch et al. 2010). Field labor occupies only the daylight, so a warmer day works on the worker, pressing effort against the physiological ceiling of sustained outdoor exertion and lowering the productivity of labor itself (Dunne, Stouffer, and John 2013; Somanathan et al. 2021). A daily average folds the two channels into one number; the analysis keeps them apart. Equivalently, entering the two margins jointly separates the daily *level* (their mean) from the diurnal *range* (their difference): the level moves the farm's scale—area, output, and self-cultivation—while the range drives the reallocation between wage labor and the seasonal margin and the harvest price, and a degree-day average retains only the level.

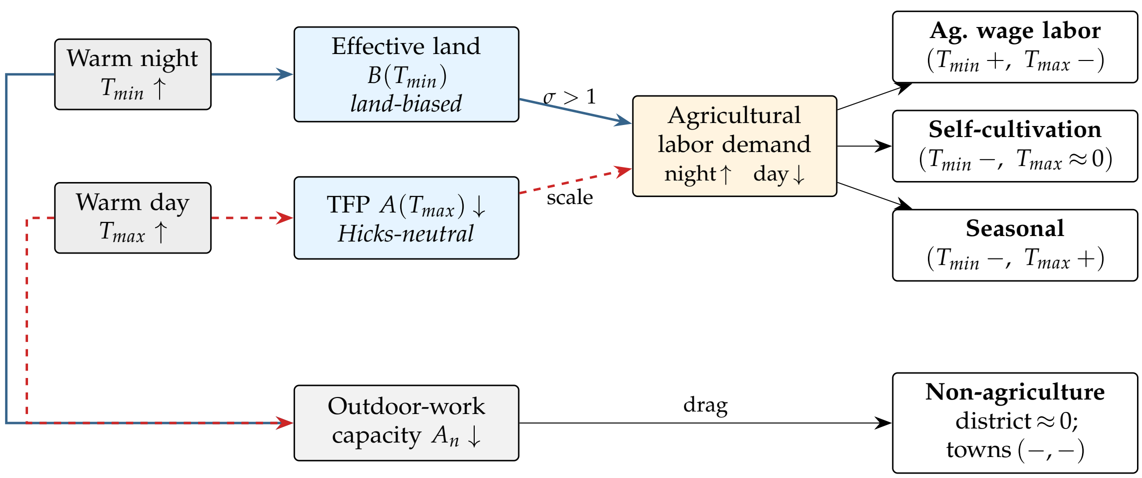
Two shocks, two factors. The framework assigns each margin to the factor it acts on: nighttime warming is a *land-biased* shock that raises or lowers the productive services of cultivated land, while daytime warming is a *Hicks-neutral* loss of total factor productivity. What makes the night margin distinctive is the factor it lands on, not the sign of its effect: the nocturnal-respiration and overnight-heat-load channels could in principle either degrade or enhance effective land, and which way is an empirical question the data resolve—humidity-adjusted nights *raise* cereal yields, the signature of net augmentation. The distinction has bite only because labor and land are gross substitutes—the farm can trade one for the other, an elasticity of substitution σ above one. When a warmer night changes the quality of an acre, a farm that substitutes readily reworks its land-to-labor mix and works the land harder, hiring *more* field labor rather than less; the direction of the labor response is set by the factor the shock lands on, not by what happens to yields. A neutral productivity loss lands on no factor in particular: it leaves the land-to-labor ratio intact and shrinks the scale of the operation, pulling down the marginal product of labor and the field wage and, where the food market is tight, raising the harvest price. The two margins therefore move farm-labor demand in *opposite* directions—warmer nights raise it, warmer days lower it. That opposition is the entire content of the decomposition, and gross substitution is what makes it possible: under Cobb–Douglas ($\sigma = 1$) a land-augmenting shock folds into total factor productivity and becomes observationally neutral, the two channels collapse, and a daily average loses nothing.

Where the labor goes: the seasonal sink. Workers divide across four states—agricultural wage labor, self-cultivation, non-agricultural work, and casual seasonal work—whose shares sum to one, so the margins move as one offsetting system, and the casual seasonal pool is the flexible residual that clears the reallocation. When nighttime warming raises the demand for farm labor, the farm hires: the wage-labor share rises and draws from the cultivators who give up their marginal plots and the casual workers who attach to the harvest—the signature of a commercializing farm that retains and reorganizes labor rather than releasing it. When daytime warming collapses the marginal product, the wage-labor market gives way and returns workers to the seasonal margin. Self-cultivation falls under both margins, as marginal plots go unworked.

The price test of market integration. The decomposition's openness is tested, not assumed. Only a productivity-driven (daytime) output loss raises the local harvest price; the nighttime contraction leaves it flat. That asymmetry is the signature of a district whose food price is set largely by trade rather than by local scarcity—partial market integration. Appendix A embeds this in a general-equilibrium closure spanning open and closed food markets and shows that it is the over-identifying price restriction, not the labor signs, that places India's districts on the integrated side; a fully closed subsistence economy would instead invert the labor orientation. Partial integration is what lets a daytime output loss raise the local harvest price, and so why only the daytime margin moves it.

Rural areas, towns, and the reading of the results. The rural–urban contrast is then one of amplitude, not of sign. The factor-bias divergence is a mechanism that operates inside the agricultural sector; its footprint in local employment shares therefore scales directly with that sector's weight in the local economy. Where agriculture is the dominant sector—rural district populations—the reallocation registers sharply, amplified further by the depth of the casual-labor margin that absorbs it. Where agriculture is a minor sector—towns—the within-agriculture divergence attenuates and the dominant sector's own heat response shows through: the shared loss of capacity for outdoor informal work contracts non-agricultural employment under both margins—the long-run restraint of non-farm demand under heat (Liu, Shamdasani, and Taraz 2023; Nath 2025)—though, as the regional and wet-bulb cuts show, this attenuation need not be exact. The framework belongs to the structural-transformation tradition of Matsuyama (1992), Gollin and Rogerson (2014), and Bustos, Caprettini, and Ponticelli (2016); Appendix A develops it in full and Table A1 collects the predicted signs, with proofs in the supplemental online appendix. Figure 1 traces each channel to the labor share it moves. These signs guide the analysis rather than serving as a structure to be estimated; the next sections test them.

FIGURE 1. The diurnal mechanism: pathways from warming to each labor margin



Solid blue marks the nighttime (T_{min}) pathway and dashed red the daytime (T_{max}) pathway. A higher T_{min} augments or degrades effective land; because labor and land substitute ($\sigma > 1$), it raises agricultural labor demand and, through the seasonal sink, lifts the wage-labor share while draining self-cultivation and casual work. A higher T_{max} is a neutral productivity loss that shrinks the scale of the operation, lowering labor demand and returning workers to the seasonal margin. Both margins also erode outdoor-work capacity (A_n), which in towns—where farm labor is too small to register the reallocation—shows through as the dominant margin, a broad contraction of non-agriculture (the day/night divergence attenuated but not erased); both additionally contract cultivated area (the common supply shock, omitted as it is not a labor share). Signs in each outcome box are the estimated (T_{min}, T_{max}) responses from the district panel.

2. Data and Empirical Approach

2.1. Data Sources

I harmonize four datasets to link climate, agricultural production and structural transformation: (1) daily gridded weather data from the IMD, (2) district agricultural data (ICRISAT), (3) district and town level tables from the Indian Census, and (4) urban household-level employment data from the NSS. These datasets span across 4 decades (Census, IMD, ICRISAT), with a shorter window from the NSS (1987, 1999, 2009).

Indian Meteorological Department. Gridded data were obtained from the Indian Meteorological Department (IMD) on air temperature (1° resolution) and precipitation (0.25° resolution) between 1951-2023. Consistent with the approach in Liu, Shamdasani, and Taraz (2023), daily observations are averaged over growing seasons of the past 10 years. The growing season marks the months of June-February, which captures cropping cycles of rice in the monsoon (kharif) and wheat in the winter (rabi) seasons. The baseline exposure measures are the growing-season means of the daily minimum and maximum surface-air temperatures, T_{min} and T_{max}, which separate nighttime from daytime warming. For comparison I also construct degree days (DDs), a scalar that rescales and sums agriculturally relevant temperatures between 8 and 32 °C (Blakeslee and Fishman 2015; Schlenker, Hanemann, and C. 2006); degree days average over the diurnal cycle and so collapse the two margins into one.

$$DDs = \sum_{d} D(\mathrm{T}_d) \text{ ; where } \mathrm{T}_d = \frac{\mathrm{T}_{min,d} + \mathrm{T}_{max,d}}{2} \tag{1}$$

$$D(\mathrm{T}) = \begin{cases} 0, & \text{if } \mathrm{T} \leq 8°\mathrm{C} \\ \mathrm{T} - 8, & \text{if } 8°\mathrm{C} < \mathrm{T} \leq 32°\mathrm{C} \\ 24, & \text{if } \mathrm{T} > 32°\mathrm{C} \end{cases} \tag{2}$$

As a humidity-adjusted robustness measure I also construct growing-season wet-bulb minimum and maximum temperatures following Stull (2011); these are derived from T_{min} and T_{max} alone—no external humidity series is used (the construction is detailed in the supplemental data appendix)—and enter the wet-bulb specifications reported there.

Census of India. The core dataset is sourced from the District Census Handbooks (DCHBs) of India. Specifically, I digitize the Primary Census Abstract (PCA) tables, covering the universe of Indian towns from 1981 to 2011. A Census town meets three criteria: (1) a minimum population of 4,000, (2) at least 75% of the male working population in non-agricultural activity, and (3) a population density above 400 persons per sq. km. I construct a panel of 2,969 towns that appear consistently across all four decades, so that the analysis reflects intensive-margin adjustments rather than the reclassification of urban areas. For comparability over time, occupations are aggregated into four mutually exclusive categories: cultivators (landowners), agricultural laborers (wage workers), non-agricultural workers, and marginal workers. The "marginal" category cannot be assigned cleanly to agriculture or non-agriculture, but it lets me track the contractual *quality* of work.

ICRISAT. The ICRISAT District Level Database (DLD) tracks agricultural inputs and production in India between 1966-present. I harmonize districts to 2011-Census consistent boundaries, constructing an annual panel of cultivated area, production, yields and harvest prices for key grains: rice, wheat, maize, sorghum, barley and millets. I construct the mean of these measures over 10 years preceding each Census round, e.g. over the years 1971-1980 for the 1981 Census. Thus, the results from the temperature regressions compare contemporaneous horizons of 10-year windows with climate data. The DLD covers staple grains only; horticulture and cash crops—which have grown rapidly and differ in labor intensity, irrigation, and market structure—lie outside it, so the agricultural-economy results speak to staple-grain production and the broader composition of Indian agriculture is left to future work.

National Sample Surveys. While the Census measures the volume of labor reallocation, it does not resolve the destination sectors within non-agriculture under its 1981 definitions. To identify where displaced labor ends up, I use three rounds of the National Sample Survey (NSS) Employment and Unemployment Schedules (1987, 1999, and 2009), harmonized by IPUMS (Ruggles et al. 2024). I group occupations into "outdoor" work (cultivation, agricultural labor, construction, mining) and "indoor" work (manufacturing, services), and restrict the sample to urban households as a benchmark to the Census town-level data. This linkage reveals the destination of heat-displaced labor—casual and seasonal work rather than stable employment in manufacturing or services.

2.2. Empirical Approach

The identification relies on two approaches: a fixed effects panel estimation (3), and a long differences approach (4) described under Dell, Jones, and Olken (2014), Burke and Emerick (2016) and Liu, Shamdasani, and Taraz (2023). Let Y_{irt} be the outcome of interest (e.g., share of workers in agriculture) for district or town i, within state r, observed at time t. $\mathbf{T}_{it}$ denotes a vector of temperature measures, either degree days $\mathbf{T} = \{DDs\}$ or the diurnal surface-air pair $\mathbf{T} = \{T_{min}, T_{max}\}$. All specifications control for total precipitation

$\mathbf{P}_{it}$ and fit a state-specific linear time trend.

$$Y_{irt} = \beta \mathbf{T}_{irt} + \phi \mathbf{P}_{irt} + \mu_i + \lambda_t + \gamma_r \cdot t + \varepsilon_{irt} \tag{3}$$

$$\Delta Y_{ir} = \beta \Delta \mathbf{T}_{ir} + \phi \Delta \mathbf{P}_{ir} + \gamma_r + \varepsilon_{ir} \tag{4}$$

For temporal parity, I construct the gridded temperature and rainfall measures as a moving average over the growing seasons of the preceding ten years. Decadal climate is the inverse-distance-squared–weighted average of IMD grid points surrounding the district (within 200 km) or town (within 100 km) centroid—radii chosen to match the distance at which the residual spatial correlation of the outcomes decays (roughly 150–200 km for districts and 100 km for towns). Because the weights fall off as the squared distance, grid points beyond roughly 100 km contribute negligibly, so the assignment is insensitive to the exact cutoff.

Equation 3 is a town-decade panel with two-way fixed effects and a state-specific linear trend; because the climate regressor is itself a ten-year moving average, I read its coefficients as the *decadal* response of structural transformation to heat. Equation 4 averages the outcome Y and covariates X over two endpoints—1981–91 and 2001–11—and regresses $\Delta Y = Y_2 - Y_1$ on $\Delta X = X_2 - X_1$. This long-differenced estimator recovers the effect of a sustained increase in $\mathbf{T}$ on the outcome trend ΔY, measured against its secular trend $\Delta \bar{Y}$ over the sample. Both estimators use the same decadally smoothed climate; the long difference adds one further decade of horizon and absorbs unobserved state-level trajectories $(\gamma_r \cdot t)$, so I read its coefficients as the longer-run, adaptation-inclusive response (Burke and Emerick 2016) rather than as a sharp short-run/long-run contrast.

I enter T_{min} and T_{max} simultaneously and read each coefficient as the partial effect of one margin holding the other—and rainfall—constant. Inference faces two distinct dependencies. The first is the decadal *persistence* of the climate regressor (a ten-year moving average), which inflates rejection rates unless the estimator accounts for within-unit serial correlation. Appendix Table A17 disciplines this with a placebo Monte Carlo—regressing the key outcomes on the real climate series randomly permuted across districts, which preserves each series' decadal persistence while breaking its link to the outcome—and I select the preferred panel estimator for equation 3 as the specification that rejects the null in roughly 5% of simulations. Unadjusted and heteroskedasticity-robust standard errors over-reject sharply, whereas clustering by district with state-specific linear trends restores the nominal 5% size. The second dependency is spatial; because the permutation reassigns whole series across units it breaks the regressor's spatial structure, so the

Monte Carlo does not speak to spatial correlation, which I address separately with Conley spatial-HAC standard errors (Conley 1999) reported alongside the clustered errors throughout; Appendix Table A7 sweeps the Conley radius to 200 km and the headline coefficients remain significant.

The simultaneous estimator addresses two identification challenges. First, because T_{min} and T_{max} are correlated ($\rho \approx 0.78$ across districts), entering either alone would induce omitted-variable bias; entering them together isolates their partial effects and identifies a narrowing versus widening of the local diurnal range. The district and decade fixed effects and state trends absorb most of this co-movement, so the *estimating* variation is close to orthogonal (within-transformed VIF ≈ 1; Appendix Table A5) and the partial signs are well-identified. By Frisch–Waugh–Lovell the coefficients load on exactly this residual variation, so it is the within correlation, not the raw ρ, that governs whether the two margins can be separated—and within, they are all but orthogonal. Second, absorbing state-specific trends accounts for time-varying state policies and compares towns within their local economic geography under similar baseline climate trajectories. Correlograms of the residuals confirm that the spatial correlation of these heat shocks in the key outcomes decays within roughly 100 km for towns and 150–200 km for districts.

3. Results

Table 1 decomposes the effects of warming nights and days on the district agricultural economy: Panel A reads down the supply chain for the major cereals—rice, wheat, maize, sorghum, barley, and millets—and Panel B reports the within-district reallocation of labor. The nighttime minimum and the daytime maximum carry opposite signs across the price and labor outcomes, consistent with the theoretical framework.

Both margins shrink the farm. The contraction in cultivated land is statistically identical across the two margins—a $1°$C rise cuts log area by 0.22 either way ($F = 0.01$)—while production falls comparably rather than identically (-0.15 at night, -0.22 by day, $F = 0.8$). I read the milder nighttime loss as labor intensification at work: the significant nighttime rise in wage labor adds output back through the production identity, cushioning the extensive-area loss, where daytime labor-shedding compounds it. The margins part on the price. The daytime maximum raises the log harvest price by 0.13 while the nighttime minimum holds it flat, a split sharp at $F = 13.3$; the field wage edges down under both (-0.12 by day, -0.04 at night, too close to separate). Daytime warming acts as a supply shock: the daytime loss of land and grain passes straight into a dearer harvest price, where the nighttime price holds flat. Cereal yields move little under either margin, the flat reading the land-augmentation channel anticipates once the gain to standing crops nets against the lost area.

Panel B locates the labor. A warmer night raises the agricultural wage-labor share by

TABLE 1. Panel Estimates of Nighttime vs Daytime Warming in Districts

Panel A. ICRISAT Districts (1981-2011)

	Log Levels (All Grains)				
	Cultivated Area	Production	Harvest Price Index	Field Wage	Yield Index
	(1)	(2)	(3)	(4)	(5)
T_{min}	-0.218***	-0.145**	-0.042	-0.038	0.051
	(0.0459)	(0.0641)	(0.0311)	(0.0603)	(0.0405)
	[0.0459]	[0.0643]	[0.0311]	[0.0599]	[0.0403]
T_{max}	-0.225***	-0.221***	0.130***	-0.119*	-0.046
	(0.0562)	(0.0637)	(0.0341)	(0.0703)	(0.0474)
	[0.0560]	[0.0639]	[0.0345]	[0.0677]	[0.0471]
Observations	1,243	1,243	1,191	1,115	1,211
Dep Var. Mean	5.45	5.77	5.66	2.86	7.14
F-stat ($T_{min} = T_{max}$)	0.01	0.80	13.34	0.74	2.63

Panel B. Census Districts (1981-2011)

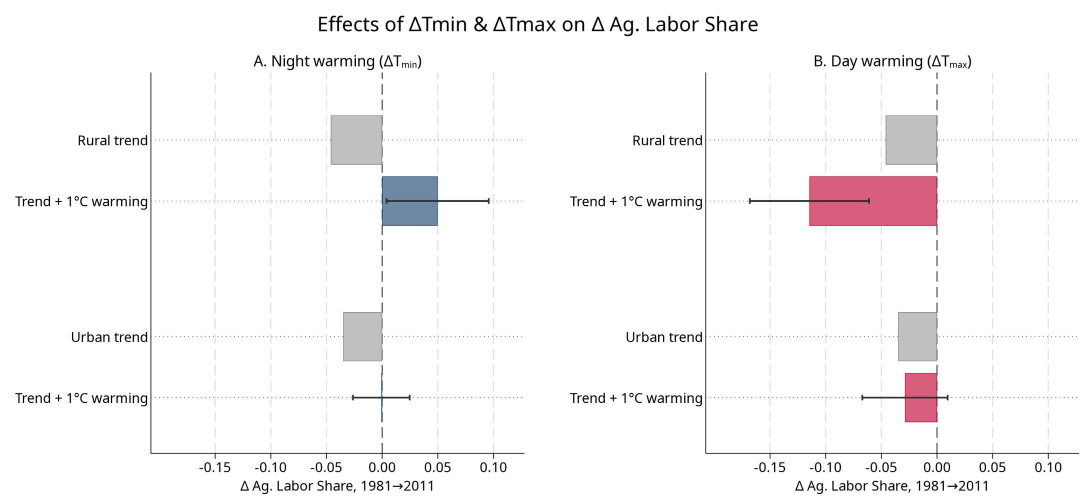
	Share of Workers				
	Ag. Labor	Cultivator	Seasonal	Non-Agriculture	Log Workers
	(1)	(2)	(3)	(4)	(5)
T_{min}	0.044***	-0.028**	-0.033***	0.017	0.175**
	(0.0096)	(0.0125)	(0.0115)	(0.0114)	(0.0881)
	[0.0095]	[0.0123]	[0.0115]	[0.0112]	[0.0867]
T_{max}	-0.026***	-0.010	0.032**	0.004	-0.017
	(0.0094)	(0.0126)	(0.0129)	(0.0127)	(0.0825)
	[0.0094]	[0.0125]	[0.0130]	[0.0125]	[0.0820]
Observations	1,187	1,187	1,187	1,187	1,187
Dep Var. Mean	0.18	0.32	0.17	0.32	13.45
F-stat ($T_{min} = T_{max}$)	24.91	0.85	14.66	0.54	2.36
Rainfall Control	Y	Y	Y	Y	Y
State Linear Trend	Y	Y	Y	Y	Y

Notes: Panel A reports log cultivated area, log production, a log harvest-price index, the log field wage, and an area-weighted yield index for the major Indian cereals (rice, wheat, barley, maize, millets, sorghum). Panel B reports district shares of total workers by category and, in the last column, the log number of district workers. T_{min} and T_{max} are the growing-season (June–February) surface-air minimum and maximum, averaged over the preceding decade; all specifications control for total precipitation over the same window. The data form a decadal panel (1981–2011) on 2011 Census-consistent district boundaries, with district and decade two-way fixed effects and a state × decade linear trend. Each coefficient reports two standard errors: district-clustered in parentheses and Conley spatial–temporal HAC (50 km, four decadal lags) in brackets. The two are nearly identical and the night/day equivalence (F) tests are unchanged under the spatial correction.

4.4 points and draws that labor off self-cultivation (-2.8) and the casual seasonal margin (-3.3); a warmer day reverses the flow, cutting the wage-labor share by 2.6 points and returning 3.2 points to the seasonal margin. The two margins separate sharply—$F = 24.9$ on wage labor, $F = 14.7$ on the seasonal share—while non-agricultural employment holds flat under both ($F = 0.5$): the district reshuffles labor within agriculture. The extensive count rises at night, a warmer minimum lifting the log number of district workers by 0.18. The appendix framework grounds the split: nighttime warming augments land, and because labor and land are gross substitutes it raises the demand for field labor directly; daytime warming is a neutral productivity loss that shrinks the operation and sheds labor to the seasonal pool.

Higher nighttimes commercialize agriculture. Though the loss in grain production is comparable to higher daytimes, the district heads off the price shock through adaptation: consolidating land and intensifying agricultural labor. Higher daytimes casualize it. Facing a comparable loss, the farm contracts instead of adapting: the wage-labor market gives way, workers fall back to the casual seasonal margin, and the supply loss passes straight through to the harvest-price spike.

FIGURE 2. Long-differenced estimates of warming on the declining agricultural wage-labor trend in districts



Notes: Long-difference estimates (1981–2011) of a 1°C rise in growing-season nighttime (T_{min}) and daytime (T_{max}) surface-air temperature on the district agricultural wage-labor share, estimated separately for the rural and urban populations of each district. Grey bars give the secular trend (the mean change in the share); each colored bar adds the warming coefficient to that trend ($\bar{y} + \hat{\beta}$), with 95% confidence intervals. A colored bar whose interval clears its grey trend bar is a statistically significant warming effect. All specifications control for the long-run change in precipitation and include state fixed effects; standard errors are clustered at the district level.

Figure 2 carries the labor decomposition to its two-decade horizon and splits each district into its rural and urban populations. The long difference averages each district's two Census endpoints—1981 and 1991 against 2001 and 2011—and differences them, stripping decade-specific noise to isolate the change in climate against the change in labor. The grey bars set the stage: the agricultural wage-labor share fell across rural and urban India over these two decades, the secular structural transformation drawing labor off the land.

Warming bends that trend, and the two margins bend it in opposite directions. A warmer night raises the rural agricultural wage-labor share by 0.096 and turns its -0.046 secular decline into a $+0.050$ rise: nighttime warming hoards labor on the farm against the structural trend. A warmer day lowers the same share by 0.068, deepening the decline to -0.114: daytime heat pushes a rural agricultural exit. The opposition is sharp and holds over the two-decade horizon.

The urban populations of the same districts barely move. A warmer night lifts the urban agricultural share by 0.034—a third of the rural response—and a warmer day leaves it flat ($+0.006$, insignificant). The reversal is a rural phenomenon; the urban populations of these districts absorb little of it. The framework reads this as amplitude (Proposition A4): the diurnal response scales with the agricultural share of local employment, deep in rural districts and thin in urban ones, so the same factor-bias force registers sharply in rural areas and faintly in the town. Table A11 reports the full rural and urban decomposition behind the figure—all four worker shares and log total workers. District populations move in the same direction as workers, with the urban population rising under nighttime warming (Table A12).

Table 2 brings the long difference to the Census town and splits it by region. The cut is geographic and predetermined—a town's state, fixed long before the sample—so it is exogenous to the temperature regressors and free of the right-hand-side conditioning that a warming-based split would induce. Panel A is the night-biased Indo-Gangetic north and northwest, Panel B the day-biased peninsular Deccan and south (Figure A2). The town response is concentrated in the south.

In the peninsular south, the town's non-agricultural share contracts under both margins (-0.128 at night, -0.078 by day)—the heat-capacity drag the framework assigns to informal outdoor work (Proposition A5). Alongside it, and *outside* what the framework predicts for towns, a warmer night raises the town agricultural wage-labor share by 11.1 points ($F = 11.5$ against the daytime coefficient) and lifts the total town workforce ($+0.54$). This nighttime pull back onto the land is large for a sector the town definition holds below a quarter of employment; the log-linear framework treats town agriculture as too small to register the reallocation and does not generate it. I read it as a robust but unmodeled empirical pattern—it survives in the decadal panel as well as the long difference—rather than a confirmation of the mechanism, and treat the southern result as suggestive, its magnitude resting on the newly digitized town panel—though the

TABLE 2. Town Labor Reallocation by Region: Indo-Gangetic North vs Peninsular South

	Δ Share of Workers				
	Ag. Labor	Cultivator	Seasonal	Non-Agriculture	Log Total Workers
	(1)	(2)	(3)	(4)	(5)
Panel A: Indo-Gangetic North (night-biased belt)					
Δ T_{min}	0.028**	-0.038**	-0.028	0.036*	0.010
	(0.0139)	(0.0154)	(0.0171)	(0.0215)	(0.1477)
	[0.0148]	[0.0156]	[0.0179]	[0.0218]	[0.1449]
Δ T_{max}	0.037*	0.005	0.005	-0.048	-0.030
	(0.0200)	(0.0204)	(0.0189)	(0.0299)	(0.1472)
	[0.0215]	[0.0216]	[0.0223]	[0.0312]	[0.1471]
Observations	1,618	1,618	1,618	1,618	1,618
F-stat ($T_{min} = T_{max}$)	0.17	4.32	2.55	8.09	0.13
Panel B: Peninsular South (day-biased belt)					
Δ T_{min}	0.111***	0.024	-0.005	-0.128***	0.544*
	(0.0198)	(0.0150)	(0.0164)	(0.0272)	(0.2903)
	[0.0199]	[0.0154]	[0.0161]	[0.0280]	[0.2668]
Δ T_{max}	0.010	0.036*	0.032	-0.078**	0.362
	(0.0247)	(0.0198)	(0.0239)	(0.0326)	(0.2842)
	[0.0266]	[0.0191]	[0.0232]	[0.0326]	[0.2691]
Observations	1,351	1,351	1,351	1,351	1,351
F-stat ($T_{min} = T_{max}$)	11.48	0.18	1.80	1.73	1.17
Rainfall Control	Y	Y	Y	Y	Y
State FE	Y	Y	Y	Y	Y

Notes: Long differences (1981–2011) of each balanced Census-town worker share on the long-run change in T_{min} and T_{max}, split by region. The split is a *predetermined, exogenous* heterogeneity cut—the town's state, not a function of the temperature regressors—so it avoids conditioning on the right-hand side that a warming-based split would entail. Panel A is the night-biased Indo-Gangetic north and northwest (Punjab, Haryana, Delhi, Uttar Pradesh, Bihar, West Bengal, Rajasthan, Gujarat, and the Himalayan states); Panel B the day-biased peninsular Deccan and south (Maharashtra, Karnataka, Andhra Pradesh, Tamil Nadu, Kerala, Madhya Pradesh, Chhattisgarh, Odisha); see Figure A2. The four mutually exclusive Census worker categories—agricultural wage labor, cultivators, seasonal, and non-agricultural—sum to one, so the share coefficients sum to approximately zero. Column 5, set apart by a gap, reports the change in log total workers. The F-statistic tests $T_{min} = T_{max}$. T_{min} and T_{max} are the growing-season (June–February) surface-air minimum and maximum; all specifications control for the long-run change in precipitation and include state fixed effects. Each coefficient reports district-clustered standard errors in parentheses and Conley 50 km spatial-HAC errors in brackets. The balanced-versus-statutory-town decomposition behind the pooled town result is in Appendix Table A14. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

labor-share shift is robust to dropping any single southern state (Appendix Tables A14 and A15).

The northern plains are comparatively quiescent. Agricultural wage labor edges up under both margins ($+0.028$, $+0.037$) and the non-agricultural share shows only a weak long-difference divergence (a nighttime $+0.036$ against a daytime -0.048, $F = 8.1$) that does not survive the decadal panel. Where the rural district reshuffles labor within agriculture, the southern town instead sheds it from the informal non-farm sector—a reallocation that stalls, rather than advances, the structural transition. The balanced-versus-statutory-town decomposition, including the in-migration that lets established towns grow their workforce even as the non-farm share contracts, is reported in Appendix Table A14.

The appendix confirms the decomposition under alternative measures (Appendix Section C): a single degree-days measure collapses the opposition into one muted coefficient (Table A4); and wet-bulb temperatures sharpen the labor split while humid nights raise cereal yields outright ($+0.15$), the land-augmenting signature ($B' > 0$) that the flat dry-bulb yield only implies (Table A8).

4. Conclusion

This paper asks which way warming pushes structural transformation. I decompose it into nighttime and daytime surface-air components and read the labor reallocations through a general-equilibrium framework in the tradition of Matsuyama (1992). The framework's content is not the labor signs it is built to reproduce but a single over-identifying restriction: only the daytime margin, not the nighttime, moves the local harvest price, and the data confirm precisely this asymmetry ($F = 13.3$ on price equality). That restriction corroborates partial market integration independently of the labor reallocation, and distinguishes the mechanism from the closed-economy "food problem" of Liu, Shamdasani, and Taraz (2023) and Nath (2025), which turns on a national, long-run subsistence price spiral. The framework guides the signs rather than supplying a structure to estimate; its content is reduced-form panel and long-difference regressions that test those statics.

The approach has limits. Collinearity might seem to manufacture the opposing signs, but in the surface-air baseline it does not: the two margins are correlated at only $\rho \approx 0.78$, and the district and decade fixed effects and state trends absorb almost all of that co-movement, leaving the estimating variation near-orthogonal (within-transformed VIF ≈ 1, condition number ≈ 1.1; Appendix Table A5). Each margin already carries its sign when it enters alone—significantly so in both the panel and the long difference—so the correlation widens the standard errors without reversing the point estimates. The qualification is the nighttime sign's robustness versus the daytime's: the night response holds under individual entry throughout, whereas under the humidity-adjusted wet-bulb long difference the daytime sign emerges only once both margins enter together. That wet-bulb specification is supplementary, not a primary pillar: the wet-bulb measure is a deterministic nonlinear transform of the same T_{min} and T_{max} (it uses no external humidity data; see the data appendix), and its range-compressing construction makes the wet-bulb pair far more collinear than the dry-bulb baseline ($\rho \approx 0.93$, within-transformed VIF ≈ 3.5; Appendix Table A5). The daytime wet-bulb sign-flip on joint entry is thus the expected symptom of that collinearity rather than independent humidity information—and it is why the surface-air baseline, near-orthogonal in its estimating variation, carries the result. Entering T_{min} and T_{max} jointly, moreover, spans the same two-dimensional regressor space as the daily mean and the diurnal range, so the decomposition is a reparameterization of mean-plus-range rather than an artifact of the chosen basis: entered directly, the diurnal range carries a significant coefficient on the labor and price margins (Appendix Table A6). The divergence further assumes labor and land are gross substitutes ($\sigma > 1$), a keystone the paper does not estimate; the framework guides the signs but is not itself tested; and the town non-agricultural drag is the dominant margin rather than a strictly uniform one—the day/night divergence resurfaces in the wet-bulb and Indo-Gangetic-North cuts. Nonlinearity and spatial sorting, held fixed by the linear framework, are left for later work.

The framework's distinctive prediction—that a closed subsistence economy would invert the pattern—is the natural next test: India's pre-1991 Censuses, before liberalization opened the agricultural economy, offer a closed-regime counterfactual to falsify the open-regime reading. The pattern itself is consistent: warmer nights hold labor on the farm without the price and wage distress that warmer days bring, while daytime heat restrains the non-agricultural transition far more strongly. Because the two margins carry opposite signs on agricultural labor, the trajectory of the diurnal temperature range matters as much as the level: regions where nights are warming faster than days—compressing the range, as in the Indo-Gangetic belt—face agricultural consolidation and commercialization; regions where days are outpacing nights and the range is expanding face the double squeeze of farm-labor casualization and stalled urban structural transformation. The urban shift into informal services it leaves is less an engine of growth than a physiologically constrained churn for survival.

References

Blakeslee, David S. and Ram Fishman. 2015. "Weather Shocks, Agriculture, and Crime: Evidence from India." 10.2139/ssrn.2428249. [Online; accessed 23. Nov. 2024].

Burke, Marshall and Kyle Emerick. 2016. "Adaptation to Climate Change: Evidence from US Agriculture." *American Economic Journal: Economic Policy* 8 (3): 106–40. 10.1257/pol.20130025.

Bustos, Paula, Bruno Caprettini, and Jacopo Ponticelli. 2016. "Agricultural Productivity and Structural Transformation: Evidence from Brazil." *American Economic Review* 106 (6): 1320–65. 10.1257/aer.20131061.

Colmer, Jonathan. 2021. "Temperature, Labor Reallocation, and Industrial Production: Evidence from India.". https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3900866. [Online; accessed 14. Jan. 2026].

Conley, Timothy G. 1999. "GMM estimation with cross sectional dependence." *Journal of Econometrics* 92 (1): 1–45. 10.1016/s0304-4076(98)00084-0.

Dell, Melissa, Benjamin F. Jones, and Benjamin A. Olken. 2014. "What Do We Learn from the Weather? The New Climate-Economy Literature." *Journal of Economic Literature* 52 (3): 740–98. 10.1257/jel.52.3.740.

Dunne, J. P., R. J. Stouffer, and J. G. John. 2013. "Reductions in labour capacity from heat stress under climate warming." *Nature Climate Change* 3 (6): 563–566. 10.1038/nclimate1827.

Gollin, Douglas and Richard Rogerson. 2014. "Productivity, transport costs and subsistence agriculture." *Journal of Development Economics* 107: 38–48. 10.1016/j.jdeveco.2013.10.007.

Herrendorf, Berthold, Richard Rogerson, and Akos Valentinyi. 2014. "Growth and Structural Transformation." In *Handbook of Economic Growth*, vol. 2, pp. 855–941. Elsevier. 10.1016/B978-0-444-53540-5.00006-9.

Jessoe, Katrina, Dale T. Manning, and J. Edward Taylor. 2017. "Climate Change and Labour Allocation in Rural Mexico: Evidence from Annual Fluctuations in Weather." *The Economic Journal* 128 (608): 230–261. 10.1111/ecoj.12448.

Liu, Maggie, Yogita Shamdasani, and Vis Taraz. 2023. "Climate Change and Labor Reallocation: Evidence from Six Decades of the Indian Census." *American Economic Journal: Economic Policy* 15 (2): 395–423. 10.1257/pol.20210129.

Mall, Rajesh Kumar, Manisha Chaturvedi, Nidhi Singh, Rajeev Bhatla, Ravi Shankar Singh, Akhilesh Gupta, and Dev Niyogi. 2021. "Evidence of asymmetric change in diurnal temperature range in recent decades over different agro-climatic zones of India." *International Journal of Climatology* 41 (4): 2597–2610. 10.1002/joc.6978.

Matsuyama, Kiminori. 1992. "Agricultural productivity, comparative advantage, and economic growth." *Journal of Economic Theory* 58 (2): 317–334. 10.1016/0022-0531(92)90057-O.

Munshi, Kaivan and Mark Rosenzweig. 2016. "Networks and Misallocation: Insurance, Migration, and the Rural-Urban Wage Gap." *American Economic Review* 106 (1): 46–98. 10.1257/aer.20131365.

Nath, Ishan. 2025. "Climate Change, the Food Problem, and the Challenge of Adaptation through Sectoral Reallocation." *Journal of Political Economy*. 10.1086/734725.

Peng, Shaobing, Jianliang Huang, John E. Sheehy, Rebecca C. Laza, Romeo M. Visperas, Xuhua Zhong, Grace S. Centeno, Gurdev S. Khush, and Kenneth G. Cassman. 2004. "Rice yields decline with higher night temperature from global warming." *Proceedings of the National Academy of Sciences* 101 (27): 9971–9975. 10.1073/pnas.0403720101.

Ruggles, Steven, Lara Cleveland, Rodrigo Lovaton, Sula Sarkar, Matthew Sobek, Derek Burk, Dan Ehrlich, Quinn Heimann, and Jane Lee. 2024. "Integrated Public Use Microdata Series, International: Version 7.5 [dataset]." https://doi.org/10.18128/D020.V7.5.

Schlenker, Wolfram, W. Michael Hanemann, and Anthony C. 2006. "The Impact of Global

Warming on U.S. Agriculture: An Econometric Analysis of Optimal Growing Conditions." *Rev. Econ. Stat.* 88 (1): 113–125. https://ideas.repec.org/a/tpr/restat/v88y2006i1p113-125.html.

Somanathan, Eswaran, Rohini Somanathan, Anant Sudarshan, and Meenu Tewari. 2021. "The Impact of Temperature on Productivity and Labor Supply: Evidence from Indian Manufacturing." *Journal of Political Economy*. 10.1086/713733.

Stull, Roland. 2011. "Wet-Bulb Temperature from Relative Humidity and Air Temperature." *Journal of Applied Meteorology and Climatology* 50 (11): 2267–2269. 10.1175/JAMC-D-11-0143.1.

Welch, Jarrod R., Jeffrey R. Vincent, Maximilian Auffhammer, Pura F. Moya, Achim Dobermann, and David Dawe. 2010. "Rice yields in tropical/subtropical Asia exhibit large but opposing sensitivities to minimum and maximum temperatures." *Proceedings of the National Academy of Sciences* 107 (33): 14562–14567. 10.1073/pnas.1001222107.

Appendix A. Theoretical Framework

Full derivations and proofs are in the supplemental online appendix; this section gives the structure and the predictions it delivers.

Primitives. A representative district produces an agricultural good (relative price p) and a non-agricultural numeraire. Agricultural output combines effective labor H_a and effective land BZ_c in a CES aggregator scaled by total factor productivity,

$$Y_a = A\big[\,\theta\, H_a^{\rho} + (1-\theta)(BZ_c)^{\rho}\,\big]^{1/\rho}, \qquad \rho \equiv \frac{\sigma-1}{\sigma}, \quad \theta \in (0,1), \tag{A1}$$

with $\sigma > 0$ the elasticity of substitution between labor and land, Z_c cultivated area, $A = A(T_{max})$ Hicks-neutral, and $B = B(T_{min})$ land-augmenting. Non-agriculture produces the numeraire under decreasing returns to labor (curvature ν) with capacity A_n, and food carries subsistence content through a demand multiplier $\mathcal{M} > 1$.

Labor identity. The workforce $\bar{L}$ divides across four states whose shares sum to one,

$$l_a + l_c + l_n + l_s = 1, \tag{A2}$$

wage labor l_a, self-cultivation l_c, non-agriculture l_n, and a casual seasonal margin l_s that absorbs the residual. The reservation distribution applies to the non-cultivating population, so market supply is $L_a + L_n = (\bar{L} - L_c)\,G(w)$ and seasonal work $L_s = (\bar{L} - L_c)[1 - G(w)]$ (elasticity η_s), leaving a positive cultivator mass $L_c > 0$ and preserving (A2). The self-cultivation margin is set by a marginal-cultivator condition—a landholder hires out only when the wage exceeds the return on an own marginal parcel, $\rho(L_c) = w$ with $\rho' < 0$—so a higher wage lowers self-cultivation, $dL_c/dw < 0$. Mobility equalizes w; with family labor on inframarginal own land held fixed at the margin, field-labor demand passes through one-for-one to the hired (census) margin, $\hat{L}_a = \hat{H}_a$. The worker shares in Table A1 are the components of (A2), so they move as one offsetting system.

Closures and shocks. A single openness parameter closes the price,

$$\hat{p} = \chi\,(\hat{Y}_n - \mathcal{M}\,\hat{Y}_a), \qquad \chi \in [0,1], \tag{A3}$$

with $\chi = 0$ the open district ($\hat{p} = 0$, a within-state price taker) and $\chi = 1$ closed. A hat is a log deviation; warming enters as the land-bias shock $b \equiv (B'/B)\,dT_{min}$, the neutral shock $a \equiv (A'/A)\,dT_{max} \leq 0$, and the capacity shock $a_n \equiv \hat{A}_n$. Let $s_Z = 1 - s_L$ be land's revenue share and $\kappa \equiv \sigma/s_Z$.

Micro-to-macro and labor demand. The district responses aggregate micro choices. The farm chooses H_a and cultivated area Z_c to maximize profit $pY_a - wH_a - c(Z_c)$ at given (p, w); its labor first-order condition $p\,\partial Y_a/\partial H_a = w$ yields the labor demand

$$\hat{H}_a = b + \kappa(\hat{p} + a - \hat{w}), \tag{A4}$$

where the night shock b enters directly through factor bias and the day shock a only through the scale term κ. Labor-market clearing pins w and the price closure pins p, mapping the primitives into the district reduced form (A5).

Assumptions. *(A1) Gross substitutes:* $\sigma > 1$. *(A2) Channel assignment:* T_{min} augments land (B'), T_{max} is a Hicks-neutral loss ($A' < 0$), and informal non-agricultural capacity falls under both margins ($a_n < 0$); the data resolve $B' > 0$. Assumption (A1) is the keystone: at $\sigma = 1$ the night shock is observationally neutral and the margins cannot diverge (Lemma A1). The Hicks-neutral assignment for T_{max} is a conservative reduced form, not a knife-edge: the supplemental appendix shows the night/day divergence is preserved if daytime heat is instead modeled as *labor-disaugmenting* (matching the physiological reading), since under $\sigma > 1$ a labor-disaugmenting shock also lowers agricultural labor demand.

Equilibrium and predictions. In the open district ($\hat{p} = 0$), demand (A4) and market clearing give

$$\hat{L}_a = \frac{\Delta_o'}{\Delta_o}\,(\kappa a + b), \qquad 0 < \frac{\Delta_o'}{\Delta_o} < 1, \tag{A5}$$

with $\Delta_o, \Delta_o' > 0$ collecting the elasticities, so the two margins carry opposite signs. The framework is built to reproduce five facts: agricultural wage labor rises with night and falls with day warming (panel, long difference, and wet-bulb); the seasonal margin moves oppositely; the harvest price rises under day warming alone; district non-agriculture is flat; and town non-agriculture falls under both margins over long differences. The lemma and propositions below state the comparative statics tested in Section 3.

LEMMA A1 (Cobb–Douglas is silent). *If $\sigma = 1$, the land-bias shock b is observationally equivalent to a neutral productivity shock, and agricultural labor cannot respond with opposite signs to the two margins.*

PROPOSITION A1 (Open-economy divergence). *Under Assumptions (A1)–(A2) with $\chi < \chi^*$, agricultural labor rises with nighttime and falls with daytime warming, $\partial \hat{L}_a / \partial T_{min} > 0$ and $\partial \hat{L}_a / \partial T_{max} < 0$. The seasonal margin moves oppositely, the field wage shares the sign of $\kappa a + b$, and non-agricultural labor moves with $-\hat{w}$.*

PROPOSITION A2 (Openness threshold). *The orientation in Proposition A1 holds if and only if $\chi < \chi^* \equiv 1/(\sigma \mathcal{M})$. Because $\sigma > 1$, the nighttime sign binds first: the night response pins down how open the economy must be.*

PROPOSITION A3 (Closed economy reverses). *In the closed economy ($\chi = 1$) with subsistence ($\mathcal{M} > 1$) and $\sigma > 1$, both signs invert—nighttime warming lowers and daytime warming raises agricultural labor—through the subsistence income effect. The observed orientation therefore places the district in the open regime.*

PROPOSITION A4 (Amplitude). *Rural and urban populations share the signs of Proposition A1; the magnitude scales with the agricultural employment share and the seasonal-supply elasticity η_s, not with the degree of closure. The rural–urban contrast is one of amplitude, not of sign.*

PROPOSITION A5 (Non-agricultural drag dominates in towns). *Where agriculture is too small to register the reallocation, $a_n < 0$ delivers a contraction of non-agricultural employment under both margins, $\partial \hat{L}_n / \partial T_{min} < 0$ and $\partial \hat{L}_n / \partial T_{max} < 0$, as the dominant term—attenuating, rather than eliminating, the day/night divergence.*

Appendix B. Summary Statistics

Appendix C. Robustness of the District Decomposition

This section reproduces the district decomposition of Table 1 under alternative climate measures, an alternative estimator, and finer geographic and sectoral cuts.

TABLE A1. Model predictions: comparative statics

Outcome	Parametric solution	Sign (T_{min}, T_{max})
Cultivated area	extensive margin, both shifters	$(-, -)$
Yield index	$a + s_L(\hat{L}_a - \hat{Z}_c) + s_Z b$	$(+, \approx 0/-)$
Production	$\hat{Y}_a = a + s_L(\hat{L}_a - \hat{Z}_c) + s_Z b$	$(-, -)$
Field wage	$\hat{w} = \phi_a(\kappa a + b)/\Delta_o$	$(\approx 0, -)$
Harvest price	$\hat{p} = \chi(\hat{Y}_n - \mathcal{M}\hat{Y}_a)$	$(\approx 0, +)$
Ag. wage labor	$\hat{L}_a = (\Delta'_o/\Delta_o)(\kappa a + b)$	$(+, -)$
Self-cultivation	night-bias residual	$(-, \approx 0)$
Seasonal	$\propto -\hat{w}$, opposes $\hat{L}_a$	$(-, +)$
District non-agriculture	$\hat{L}_n = -\hat{w}/(1-\nu)$, a_n=0	$(\approx 0, \approx 0/+)$
Town non-agriculture	$\hat{L}_n = (a_n - \hat{w})/(1-\nu)$, a_n<0	$(-, -)$
Closed-economy counterfactual	χ=1 in the general solution	$(-, +)$ reversed

Notes: Signs are the model's predicted responses of each outcome to nighttime (T_{min}) and daytime (T_{max}) warming, holding the other margin fixed; ≈ 0 marks a response the model leaves weak or indeterminate. The district non-agricultural entry follows the wage-driven schedule $\hat{L}_n = -\hat{w}/(1-\nu)$: nighttime warming leaves it near zero (the field wage barely moves), while daytime warming implies a *weak positive* response ($\hat{w} < 0$), which the deep, elastic seasonal margin absorbing the shed labor damps to a magnitude empirically indistinguishable from zero (Table 1, $F = 0.5$)—hence the $(\approx 0, \approx 0/+)$ entry. $\kappa = \sigma/s_Z$; s_L, s_Z are the labor and land cost shares; ϕ_a is agriculture's weight in labor demand; $\hat{Z}_c$ is the change in cultivated land; and $\Delta_o, \Delta'_o > 0$ collect the supply and demand elasticities. The closed-economy row is the counterfactual the data reject (Proposition A3). The production sign relies on the area contraction dominating the intensive gain in $s_L(\hat{L}_a - \hat{Z}_c) + s_Z b$; the yield row is weakly positive at night, so the flat dry-bulb yield estimate is an extensive-margin artifact (the wet-bulb yield is significantly positive). The static framework fixes the workforce $\bar{L}$ to isolate share reallocation and so makes no prediction for total workers; the empirical log-workers response operates through participation and migration, outside the static model.

TABLE A2. Town Summary Statistics

	1981	1991	2001	2011
Panel A: All Towns				
Town Characteristics				
Population	42,322.64	65,209.97	74,454.70	71,033.23
Total Workers	12,786.26	19,651.29	23,853.15	25,122.46
Area (sq. km)	14.54	16.18	17.13	14.01
Share of Workers by Category				
Non-Agricultural	0.73	0.74	0.74	0.72
Agricultural Labor	0.12	0.12	0.07	0.08
Cultivation	0.11	0.10	0.06	0.05
Seasonal	0.04	0.04	0.13	0.16
Climatic Indicators (Growing Seasons, 10y MA)				
Degree Days (DDs)	16.00	16.16	16.26	16.53
Daily Temp (° C)	24.07	24.23	24.32	24.58
Nighttime: T_{min} (° C)	18.28	18.46	18.69	19.01
Daytime: T_{max} (° C)	29.85	30.00	29.95	30.15
Daily Precipitation (mm.)	0.98	1.00	0.98	1.01
Observations	3,714	4,402	4,857	7,592
Panel B: Balanced Panel (1981–2011)				
Town Characteristics				
Population	45,259.05	82,584.48	105560.07	131999.55
Total Workers	13,637.92	24,691.57	33,472.11	46,110.79
Area (sq. km)	15.05	18.60	20.55	20.25
Share of Workers by Category				
Non-Agricultural	0.74	0.74	0.76	0.74
Agricultural Labor	0.11	0.12	0.06	0.07
Cultivation	0.12	0.10	0.05	0.05
Seasonal	0.04	0.03	0.12	0.15
Climatic Indicators (Growing Seasons, 10y MA)				
Degree Days (DDs)	15.82	15.96	15.98	16.20
Daily Temp (° C)	23.89	24.03	24.05	24.27
Nighttime: T_{min} (° C)	18.02	18.15	18.28	18.47
Daytime: T_{max} (° C)	29.76	29.92	29.83	30.07
Daily Precipitation (mm.)	0.98	0.97	0.96	0.95
Observations	2,969	2,969	2,969	2,969

Notes: Decadal means for the town panel, computed on the same data the town regressions use (digitized District Census Handbooks, 1981–2011, linked to IMD climate). Panel A covers all towns observed in each Census year (the panel sample); Panel B restricts to the balanced panel of 2,969 towns present in every decade (the long-difference sample). Worker shares are shares of total town workers. Climatic indicators are 10-year moving averages over the growing season—the regressors entering the estimates.

TABLE A3. District Summary Statistics

	1981	1991	2001	2011
Panel A: Rural				
Characteristics				
Population (in Thousands)	1,459.35	1,781.56	1,554.22	1,662.11
Total Workers (in Thousands)	574.58	725.07	662.23	713.37
Share of Workers by Category				
Non-Agricultural	0.19	0.20	0.24	0.23
Agricultural Labor	0.23	0.25	0.17	0.21
Cultivation	0.47	0.44	0.34	0.28
Seasonal	0.11	0.12	0.26	0.28
Climatic Indicators (Growing Seasons, 10y MA)				
Degree Days (DDs)	15.60	15.80	15.74	15.97
Daily Temp (° C)	23.66	23.87	23.80	24.03
Nighttime: T_{min} (° C)	17.73	17.95	17.97	18.15
Daytime: T_{max} (° C)	29.59	29.78	29.63	29.91
Daily Precipitation (mm.)	0.97	0.98	0.97	0.95
Observations	291	293	297	296
Panel B: Urban				
Characteristics				
Population (in Thousands)	462.83	622.66	631.67	812.01
Total Workers (in Thousands)	138.03	187.63	203.32	287.46
Town Counts				
Census Towns	7.58	8.93	10.43	17.51
Statutory Towns	5.33	6.24	4.90	5.72
Share of Workers by Category				
Non-Agricultural	0.81	0.81	0.81	0.77
Agricultural Labor	0.07	0.08	0.04	0.05
Cultivation	0.08	0.07	0.04	0.04
Seasonal	0.03	0.03	0.11	0.14
Climatic Indicators (Growing Seasons, 10y MA)				
Degree Days (DDs)	15.66	15.86	15.80	16.03
Daily Temp (° C)	23.72	23.93	23.87	24.10
Nighttime: T_{min} (° C)	17.80	18.02	18.04	18.22
Daytime: T_{max} (° C)	29.65	29.84	29.69	29.97
Daily Precipitation (mm.)	0.96	0.98	0.97	0.95
Observations	289	293	297	297

Notes: Decadal means for the district panel, computed on the same data the district decomposition uses (digitized District Census Handbooks, 1981–2011, linked to IMD climate), split into the rural (Panel A) and urban (Panel B) sectors the decomposition estimates separately. Worker shares are shares of total sector workers. Climatic indicators are 10-year moving averages over the growing season—the regressors entering the estimates.

TABLE A4. Degree-Days Specification

Panel A. ICRISAT Districts (1981-2011)				
	Log Levels (All Grains)			
	Cultivated Area	Yield Index	Field Wage	Harvest Price Index
	(1)	(2)	(3)	(4)
Δ Degree Days	-0.427***	0.014	-0.108	0.064
	(0.0872)	(0.0662)	(0.0912)	(0.0476)
	[0.0871]	[0.0656]	[0.0895]	[0.0486]
Observations	1,243	1,211	1,115	1,191
Dep Var. Mean	5.45	7.14	2.86	5.66
Panel B. Census Districts (1981-2011)				
	Share of Workers			
	Seasonal	Ag. Cultivation	Ag. Labor	Non-Agriculture
	(1)	(2)	(3)	(4)
Δ Degree Days	-0.020	-0.041**	0.036***	0.025
	(0.0182)	(0.0178)	(0.0138)	(0.0178)
	[0.0186]	[0.0176]	[0.0136]	[0.0175]
Observations	1,187	1,187	1,187	1,187
Dep Var. Mean	0.17	0.32	0.18	0.32
Rainfall Control	Y	Y	Y	Y
State Linear Trend	Y	Y	Y	Y

Notes: Replicates Table 1 replacing the separate T_{min} and T_{max} terms with a single growing-season degree-days measure (*dds_gs*), averaged over the preceding decade. Decadal panel (1981–2011) with district and decade fixed effects and a state × decade linear trend; all specifications control for precipitation over the same window. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A5. Multicollinearity Diagnostics: Surface-Air vs Wet-Bulb

	Surface-air (T_{min}, T_{max})		Wet-bulb (T_{wmin}, T_{wmax})	
	Raw	Within	Raw	Within
Correlation ρ	0.78	-0.09	0.97	0.84
Variance-inflation factor	3.3	1.0	21.6	3.5
Condition number	3.3	1.1	9.2	3.4

Notes: Pairwise correlation, variance-inflation factor (the larger margin's), and condition number for the temperature pair entering the district decomposition—the surface-air baseline (T_{min}, T_{max}) and the wet-bulb pair (T_{wmin}, T_{wmax})—each raw (cross-sectional levels) and after the baseline within transformation (district and decade fixed effects and a state-specific linear trend; rainfall included). The surface-air baseline is near-orthogonal in its estimating variation—the fixed effects and trends absorb the raw co-movement, leaving a within-transformed VIF near 1—so the partial day/night signs are not a collinearity artifact. The wet-bulb pair, whose range-compressing construction raises its correlation, remains collinear even within transformation, which is why its daytime coefficient is the fragile one; the paper leans on the surface-air baseline and treats wet-bulb as a supplementary check.

TABLE A6. Diurnal Mean and Range Decomposition

	Cultivated Area	Production	Harvest Price Index	Field Wage	Yield Index
Panel A. ICRISAT Districts (1981-2011)					
	(1)	(2)	(3)	(4)	(5)
Diurnal Mean	-0.444***	-0.366***	0.088*	-0.157*	0.006
	(0.0848)	(0.0950)	(0.0453)	(0.0898)	(0.0647)
Diurnal Range	-0.003	-0.038	0.086***	-0.041	-0.049
	(0.0289)	(0.0427)	(0.0236)	(0.0477)	(0.0300)
Observations	1,243	1,243	1,191	1,115	1,211
Dep Var. Mean	5.45	5.77	5.66	2.86	7.14

	Ag. Labor	Cultivator	Seasonal	Non-Agriculture	Log Workers
Panel B. Census Districts (1981-2011)					
	(1)	(2)	(3)	(4)	(5)
Diurnal Mean	0.018	-0.038**	-0.001	0.021	0.159
	(0.0128)	(0.0169)	(0.0178)	(0.0171)	(0.1162)
Diurnal Range	-0.035***	0.009	0.032***	-0.006	-0.096
	(0.0070)	(0.0092)	(0.0084)	(0.0085)	(0.0625)
Observations	1,187	1,187	1,187	1,187	1,187
Dep Var. Mean	0.18	0.32	0.17	0.32	13.45
Rainfall Control	Y	Y	Y	Y	Y
State Linear Trend	Y	Y	Y	Y	Y

Notes: Reparameterizes the baseline decomposition of Table 1 from the (T_{min}, T_{max}) basis into the growing-season diurnal *mean* $\frac{1}{2}(T_{min} + T_{max})$ and diurnal *range* $(T_{max} - T_{min})$. The transformation is exact and linear, so the mean coefficient equals $\beta_{T_{min}} + \beta_{T_{max}}$ and the range coefficient equals $\frac{1}{2}(\beta_{T_{max}} - \beta_{T_{min}})$; the significance of the range coefficient is identical to the night/day equality test in Table 1. The diurnal *range* carries the labor reallocation (agricultural wage labor and the seasonal margin) and the harvest-price response, while the common cultivated-area contraction loads on the *mean*—so the decomposition is a property of the range, not an artifact of the chosen basis. Decadal panel (1981–2011), district and decade fixed effects and a state $\times$ decade linear trend, precipitation control; district-clustered standard errors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A7. Conley Spatial-HAC Standard Errors: Radius Sweep

	Coefficient	Standard error by estimator				
		Cluster	50 km	100 km	150 km	200 km
Panel A. ICRISAT Districts						
Cultivated area						
T_{min}	-0.218***	(0.0459)	(0.0459)	(0.0524)	(0.0578)	(0.0611)
T_{max}	-0.225***	(0.0562)	(0.0560)	(0.0630)	(0.0684)	(0.0689)
Harvest price index						
T_{min}	-0.042	(0.0311)	(0.0311)	(0.0359)	(0.0396)	(0.0446)
T_{max}	0.130***	(0.0341)	(0.0345)	(0.0361)	(0.0392)	(0.0430)
Field wage						
T_{min}	-0.038	(0.0603)	(0.0599)	(0.0689)	(0.0792)	(0.0857)
T_{max}	-0.119*	(0.0703)	(0.0677)	(0.0795)	(0.0898)	(0.0971)
Panel B. Census Districts						
Ag. wage labor share						
T_{min}	0.044***	(0.0096)	(0.0095)	(0.0106)	(0.0115)	(0.0129)
T_{max}	-0.026***	(0.0094)	(0.0094)	(0.0115)	(0.0127)	(0.0141)
Seasonal share						
T_{min}	-0.033***	(0.0115)	(0.0115)	(0.0141)	(0.0165)	(0.0187)
T_{max}	0.032**	(0.0129)	(0.0130)	(0.0150)	(0.0173)	(0.0178)
Non-agriculture share						
T_{min}	0.017	(0.0114)	(0.0112)	(0.0114)	(0.0111)	(0.0102)
T_{max}	0.004	(0.0127)	(0.0125)	(0.0131)	(0.0129)	(0.0129)

Notes: Point estimates from the baseline decadal panel (Table 1); the standard-error columns sweep the Conley spatial–temporal HAC radius from 50 to 200 km (four decadal time lags throughout), spanning the residual spatial-correlation decay distance for districts. Standard errors rise with the radius, but the headline coefficients—the night/day split in agricultural wage labor and the seasonal margin, and the daytime harvest-price response—remain significant at the widest (200 km) radius. The coefficient column carries district-clustered significance stars. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A8. Wet-Bulb Temperatures

	Cultivated Area	Yield Index	Field Wage	Harvest Price Index
Panel A. ICRISAT Districts (1981-2011)				
	Log Levels (All Grains)			
	(1)	(2)	(3)	(4)
$\Delta\ T^w_{min}$	0.090	0.153**	0.137	-0.237***
	(0.0781)	(0.0773)	(0.1243)	(0.0598)
	[0.0778]	[0.0770]	[0.1204]	[0.0598]
$\Delta\ T^w_{max}$	-0.579***	-0.194	-0.328*	0.366***
	(0.1534)	(0.1280)	(0.1928)	(0.0936)
	[0.1529]	[0.1270]	[0.1854]	[0.0946]
Observations	1,243	1,211	1,115	1,191
Dep Var. Mean	5.45	7.14	2.86	5.66
F-stat ($T^w_{min} = T^w_{max}$)	9.01	3.07	2.28	16.59
Panel B. Census Districts (1981-2011)				
	Share of Workers			
	Seasonal	Ag. Cultivation	Ag. Labor	Non-Agriculture
	(1)	(2)	(3)	(4)
$\Delta\ T^w_{min}$	-0.083***	-0.013	0.086***	0.011
	(0.0216)	(0.0237)	(0.0173)	(0.0221)
	[0.0214]	[0.0235]	[0.0171]	[0.0217]
$\Delta\ T^w_{max}$	0.095***	-0.028	-0.079***	0.009
	(0.0349)	(0.0348)	(0.0253)	(0.0355)
	[0.0352]	[0.0344]	[0.0252]	[0.0351]
Observations	1,187	1,187	1,187	1,187
Dep Var. Mean	0.17	0.32	0.18	0.32
F-stat ($T^w_{min} = T^w_{max}$)	10.68	0.08	16.27	0.00
Rainfall Control	Y	Y	Y	Y
State Linear Trend	Y	Y	Y	Y

Notes: Replicates Table 1 using growing-season wet-bulb minimum and maximum temperatures (T^w_{min}, T^w_{max}) in place of the surface-air measures, averaged over the preceding decade. Decadal panel (1981–2011) with district and decade fixed effects and a state × decade linear trend, and precipitation controls. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A9. Wet-Bulb Temperatures: Long-Difference Labor Reallocation

	Δ Share of Workers					
	Seasonal		Ag. Labor		Non-Agriculture	
	District	Town	District	Town	District	Town
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta\ T^{w}_{min}$	-0.147***	-0.042**	0.159***	0.003	0.001	0.087***
	(0.0440)	(0.0209)	(0.0329)	(0.0225)	(0.0469)	(0.0308)
	[0.0430]	[0.0231]	[0.0329]	[0.0242]	[0.0451]	[0.0331]
$\Delta\ T^{w}_{max}$	0.144*	0.051	-0.165**	0.119***	0.049	-0.249***
	(0.0863)	(0.0390)	(0.0652)	(0.0419)	(0.0968)	(0.0612)
	[0.0848]	[0.0432]	[0.0649]	[0.0463]	[0.0936]	[0.0648]
Observations	293	2,969	293	2,969	293	2,969
Dep Var. Mean	0.14	0.10	-0.05	-0.05	0.05	0.01
F-stat ($T^{w}_{min} = T^{w}_{max}$)	5.42	2.63	11.95	3.47	0.12	14.29
Rainfall Control	Y	Y	Y	Y	Y	Y
State FE	Y	Y	Y	Y	Y	Y

Notes: Replicates Table 2 (single long difference, 1981–2011, state fixed effects) using growing-season wet-bulb minimum and maximum temperatures in place of the surface-air measures. Under humidity adjustment the town non-agricultural share diverges—warmer wet-bulb nights *raise* it and hotter wet-bulb days lower it—in contrast to the common non-agricultural drag that dominates under dry-bulb temperatures in Table 2; see Section 3. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A10. Long-Difference Estimates Within Districts

Panel A. ICRISAT Districts (Long Difference, 1981-2011)				
	Δ Log Levels (All Grains)			
	Cultivated Area	Yield Index	Field Wage	Harvest Price Index
	(1)	(2)	(3)	(4)
Δ T_{min}	-0.584***	0.052	-0.193*	-0.045
	(0.1654)	(0.1362)	(0.1041)	(0.0493)
	[0.1599]	[0.1320]	[0.1048]	[0.0515]
Δ T_{max}	-0.607***	-0.178	-0.212*	0.061
	(0.2204)	(0.1819)	(0.1227)	(0.0750)
	[0.2137]	[0.1758]	[0.1151]	[0.0730]
Observations	311	303	259	294
Dep Var. Mean	-0.08	0.37	1.92	1.36
F-stat ($T_{min} = T_{max}$)	0.03	4.56	0.02	1.81
Panel B. Census Districts (Long Difference, 1981-2011)				
	Δ Share of Workers			
	Seasonal	Ag. Cultivation	Ag. Labor	Non-Agriculture
	(1)	(2)	(3)	(4)
Δ T_{min}	-0.074***	-0.024	0.075***	0.026
	(0.0269)	(0.0308)	(0.0209)	(0.0281)
	[0.0254]	[0.0295]	[0.0202]	[0.0271]
Δ T_{max}	0.049	-0.008	-0.062**	0.020
	(0.0334)	(0.0376)	(0.0246)	(0.0365)
	[0.0327]	[0.0365]	[0.0247]	[0.0353]
Observations	293	293	293	293
Dep Var. Mean	0.14	-0.14	-0.05	0.05
F-stat ($T_{min} = T_{max}$)	12.34	0.16	26.07	0.03
Rainfall Control	Y	Y	Y	Y
State FE	Y	Y	Y	Y

Notes: Replicates Table 1 using a single long difference (1981–2011) of each outcome on the long-run change in T_{min} and T_{max}, with state fixed effects and precipitation controls, in place of the decadal panel. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A11. District Worker Shares, Long Difference (Rural vs Urban)

Panel A. Rural Workers (long difference, 1981–2011)

	Δ Share of Workers				
	Seasonal	Ag. Cultivation	Ag. Labor	Non-Agriculture	Log Total Workers
	(1)	(2)	(3)	(4)	(5)
$\Delta\ T_{min}$	-0.064**	-0.035	0.096***	0.007	0.211
	(0.0285)	(0.0314)	(0.0233)	(0.0200)	(0.2306)
	[0.0268]	[0.0299]	[0.0225]	[0.0191]	[0.2207]
$\Delta\ T_{max}$	0.056	0.011	-0.068**	-0.001	0.025
	(0.0355)	(0.0406)	(0.0272)	(0.0291)	(0.2589)
	[0.0345]	[0.0395]	[0.0277]	[0.0283]	[0.2493]
Observations	291	291	291	291	293
Dep Var. Mean	0.15	-0.15	-0.05	0.04	0.06
F-stat ($T_{min} = T_{max}$)	11.36	1.24	30.87	0.07	0.48

Panel B. Urban Workers (long difference, 1981–2011)

	Δ Share of Workers				
	Seasonal	Ag. Cultivation	Ag. Labor	Non-Agriculture	Log Total Workers
	(1)	(2)	(3)	(4)	(5)
$\Delta\ T_{min}$	-0.028**	-0.013	0.034***	0.007	0.632*
	(0.0134)	(0.0142)	(0.0130)	(0.0270)	(0.3606)
	[0.0128]	[0.0135]	[0.0125]	[0.0258]	[0.3443]
$\Delta\ T_{max}$	0.006	0.015	0.006	-0.028	0.305
	(0.0163)	(0.0165)	(0.0195)	(0.0378)	(0.3834)
	[0.0158]	[0.0155]	[0.0188]	[0.0361]	[0.3689]
Observations	291	291	291	291	293
Dep Var. Mean	0.10	-0.04	-0.03	-0.02	0.31
F-stat ($T_{min} = T_{max}$)	3.23	1.69	1.89	0.69	0.60
Rainfall Control	Y	Y	Y	Y	Y
State FE	Y	Y	Y	Y	Y

Notes: Decomposes the district worker-share results of Table 1 (Panel B) by place of residence over a single long difference (1981–2011): Panel A reports the change in shares among rural workers and Panel B among urban workers. The four mutually exclusive Census categories are exhaustive and sum to one, so the share coefficients sum to approximately zero; column 5, set apart by a gap, reports the change in log total workers (the extensive margin). All specifications include state fixed effects and control for the long-run change in precipitation. The F-statistic tests $T_{min} = T_{max}$. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A12. District Log Population (Total, Rural, Urban)

Panel A. Decadal Panel (1981–2011)			
	Δ Log Population		
	Total	Rural	Urban
	(1)	(2)	(3)
$\Delta\ T_{min}$	0.147*	0.132	0.303**
	(0.0881)	(0.0856)	(0.1494)
	[0.0867]	[0.0841]	[0.1468]
$\Delta\ T_{max}$	-0.032	-0.004	0.025
	(0.0851)	(0.0831)	(0.1650)
	[0.0843]	[0.0814]	[0.1613]
Observations	1,187	1,187	1,187
F-stat ($T_{min} = T_{max}$)	1.95	1.28	1.67
Panel B. Long Difference (1981–2011)			
	Δ Log Population		
	Total	Rural	Urban
	(1)	(2)	(3)
$\Delta\ T_{min}$	0.236	0.127	0.650*
	(0.2445)	(0.2344)	(0.3610)
	[0.2351]	[0.2245]	[0.3433]
$\Delta\ T_{max}$	0.084	0.080	0.313
	(0.2766)	(0.2688)	(0.3872)
	[0.2671]	[0.2590]	[0.3716]
Observations	293	293	293
F-stat ($T_{min} = T_{max}$)	0.28	0.03	0.64
Rainfall Control	Y	Y	Y
State FE / Trend	Y	Y	Y

Notes: Estimates of a 1°C rise in growing-season nighttime (T_{min}) and daytime (T_{max}) temperature on log district population, for the total, rural, and urban populations. Panel A is the decadal panel (1981–2011) with district and decade fixed effects and a state × decade linear trend; Panel B is the single long difference (1981–2011) with state fixed effects. All specifications control for precipitation. The F-statistic tests $T_{min} = T_{max}$. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A13. Town-Level Worker Shares

	Share of Workers			
Panel A. Census Towns – Decadal Panel (1981-2011)				
	Seasonal	Ag. Cultivation	Ag. Labor	Non-Agriculture
	(1)	(2)	(3)	(4)
Δ T_{min}	0.002	-0.008**	0.017***	-0.012
	(0.0063)	(0.0042)	(0.0052)	(0.0084)
	[0.0061]	[0.0044]	[0.0051]	[0.0082]
Δ T_{max}	0.016**	0.004	-0.007	-0.012
	(0.0069)	(0.0046)	(0.0069)	(0.0094)
	[0.0065]	[0.0044]	[0.0060]	[0.0086]
Observations	16,215	16,215	16,215	16,215
Dep Var. Mean	0.10	0.08	0.09	0.73
F-stat ($T_{min} = T_{max}$)	2.58	2.83	7.22	0.00
Panel B. Census Towns – Long Difference (1981-2011)				
	Δ Share of Workers			
	Seasonal	Ag. Cultivation	Ag. Labor	Non-Agriculture
	(1)	(2)	(3)	(4)
Δ T_{min}	-0.016	-0.009	0.063***	-0.038**
	(0.0119)	(0.0108)	(0.0116)	(0.0176)
	[0.0123]	[0.0111]	[0.0123]	[0.0187]
Δ T_{max}	0.017	0.022	0.041***	-0.078***
	(0.0144)	(0.0144)	(0.0157)	(0.0225)
	[0.0162]	[0.0148]	[0.0172]	[0.0239]
Observations	2,969	2,969	2,969	2,969
Dep Var. Mean	0.10	-0.06	-0.05	0.01
F-stat ($T_{min} = T_{max}$)	4.06	3.38	1.61	2.78
Rainfall Control	Y	Y	Y	Y
State FE / Trend	Y	Y	Y	Y

Notes: Reproduces the worker-share decomposition at the Census town level. Panel A reports decadal-panel estimates (town and decade fixed effects and a state $\times$ decade linear trend); Panel B reports the long difference (1981–2011, state fixed effects). All specifications control for precipitation. Standard errors clustered by town in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A14. Town Labor Reallocation: Balanced vs Statutory Towns

	Δ Share of Workers				
	Ag. Labor	Cultivator	Seasonal	Non-Agriculture	Log Total Workers
	(1)	(2)	(3)	(4)	(5)
Panel A: Balanced Census Towns					
Δ T_{min}	0.063***	-0.009	-0.016	-0.038**	0.272
	(0.0116)	(0.0108)	(0.0119)	(0.0176)	(0.1693)
	[0.0123]	[0.0111]	[0.0123]	[0.0187]	[0.1599]
Δ T_{max}	0.041***	0.022	0.017	-0.078***	0.241
	(0.0157)	(0.0144)	(0.0144)	(0.0225)	(0.1706)
	[0.0172]	[0.0148]	[0.0162]	[0.0239]	[0.1649]
Observations	2,969	2,969	2,969	2,969	2,969
Dep Var. Mean	-0.05	-0.06	0.10	0.01	0.55
F-stat ($T_{min} = T_{max}$)	1.61	3.38	4.06	2.78	0.11
Panel B: Statutory Towns (Municipal in 1981)					
Δ T_{min}	0.053***	-0.001	0.005	-0.061***	0.436**
	(0.0122)	(0.0128)	(0.0118)	(0.0189)	(0.1866)
	[0.0129]	[0.0130]	[0.0121]	[0.0204]	[0.1846]
Δ T_{max}	0.031*	0.022	0.035**	-0.087***	0.377**
	(0.0177)	(0.0164)	(0.0165)	(0.0244)	(0.1878)
	[0.0194]	[0.0172]	[0.0194]	[0.0259]	[0.1875]
Observations	2,159	2,159	2,159	2,159	2,159
Dep Var. Mean	-0.05	-0.06	0.10	0.01	0.56
F-stat ($T_{min} = T_{max}$)	1.45	1.78	2.74	1.02	0.35
Rainfall Control	Y	Y	Y	Y	Y
State FE	Y	Y	Y	Y	Y

Notes: Long differences (1981–2011) of each Census-town worker share on the long-run change in T_{min} and T_{max}, with state fixed effects and a precipitation control, pooling all towns rather than splitting by region (Table 2). Panel A is the balanced set of Census towns (urban in every census 1981–2011); Panel B restricts to statutory towns—those holding municipal status as of the 1981 census, the established urban core. In the pooled balanced panel both margins depress the non-agricultural share (-0.038 at night, -0.078 by day) with little day/night divergence ($F = 1.6$ on agricultural wage labor); the statutory towns sharpen the non-agricultural drag (-0.061, -0.087) yet grow their total workforce under both margins ($+0.44$, $+0.38$)—the footprint of in-migration into towns with deeper labor markets, an adaptation-through-mobility channel the framework brackets rather than generates. The four mutually exclusive Census categories sum to one; column 5, set apart by a gap, is log total workers. Each coefficient reports district-clustered SEs in parentheses and Conley 50 km spatial SEs in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A15. Peninsular-South Town Result: Leave-One-State-Out

	Ag. Wage Labor (1)	Non-Agriculture (2)	Log Total Workers (3)	Towns
Full peninsular South	0.111***	-0.128***	0.544*	1,351
	(0.020)	(0.027)	(0.290)	
– Odisha	0.130***	-0.152***	0.548	1,246
	(0.023)	(0.032)	(0.352)	
– Chhattisgarh	0.113***	-0.129***	0.573*	1,305
	(0.021)	(0.028)	(0.307)	
– Madhya Pradesh	0.104***	-0.135***	0.563	1,102
	(0.025)	(0.031)	(0.374)	
– Maharashtra	0.111***	-0.114***	0.269	1,080
	(0.021)	(0.030)	(0.195)	
– Andhra Pradesh	0.127***	-0.150***	0.664**	1,205
	(0.021)	(0.029)	(0.331)	
– Karnataka	0.105***	-0.120***	0.509*	1,142
	(0.021)	(0.028)	(0.297)	
– Kerala	0.091***	-0.114***	0.686**	1,265
	(0.018)	(0.028)	(0.285)	
– Tamil Nadu	0.107***	-0.113***	0.495*	1,112
	(0.019)	(0.026)	(0.292)	

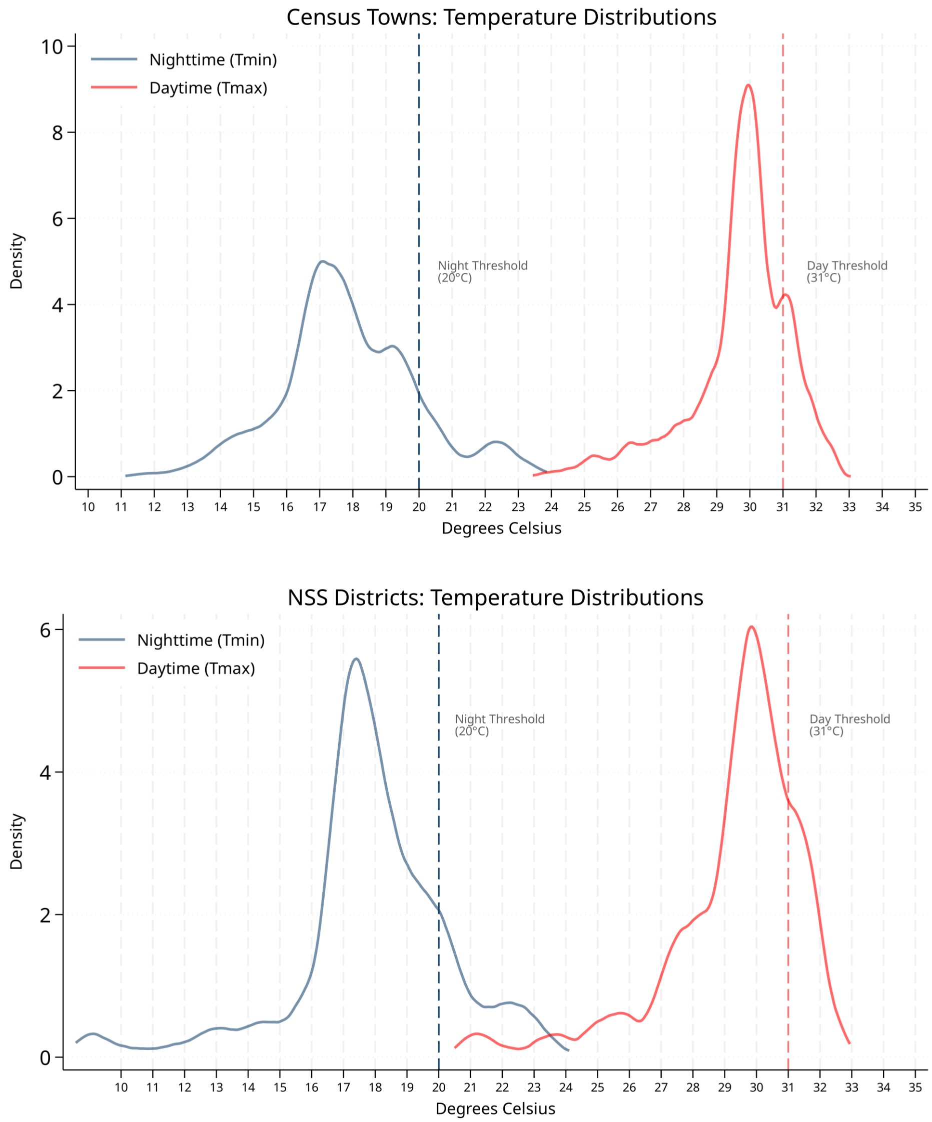
Notes: Long-difference (1981–2011) nighttime (T_{min}) coefficient on the balanced peninsular-south Census-town sample, dropping each southern state in turn, to test whether the headline southern result is driven by a single state. The agricultural wage-labor rise (+0.11) and the non-agricultural contraction (−0.13) are stable across every leave-one-out sample (all significant at $p < 0.01$); the total-workforce response is more state-sensitive. T_{min} and T_{max} enter jointly with a precipitation control and state fixed effects; district-clustered standard errors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A16. Urban Worker Shares (NSS/IPUMS), Census-Comparable

	Share of Workers					
			of which non-ag:			
	Cultivation	Ag. Labour	Informal	Formal	Log Ag-Labour Wage	Log Total Workers
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta\ T_{min}$	-0.045	-0.010	0.048**	0.007	0.366	-0.024
	(0.0275)	(0.0236)	(0.0231)	(0.0156)	(0.3052)	(0.0593)
	[0.0273]	[0.0230]	[0.0231]	[0.0153]	[0.2048]	[0.0583]
$\Delta\ T_{max}$	0.024	-0.043	0.005	0.014	-0.321	0.156
	(0.0458)	(0.0283)	(0.0300)	(0.0315)	(0.4398)	(0.1388)
	[0.0461]	[0.0285]	[0.0299]	[0.0315]	[0.2355]	[0.1388]
Observations	1,228	1,228	1,228	1,228	704	1,228
Dep Var. Mean	0.39	0.21	0.21	0.19	5.36	13.08
F-stat ($T_{min} = T_{max}$)	1.99	1.06	2.03	0.04	2.14	1.58
Rainfall Control	Y	Y	Y	Y	Y	Y
State Linear Trend	Y	Y	Y	Y	Y	Y

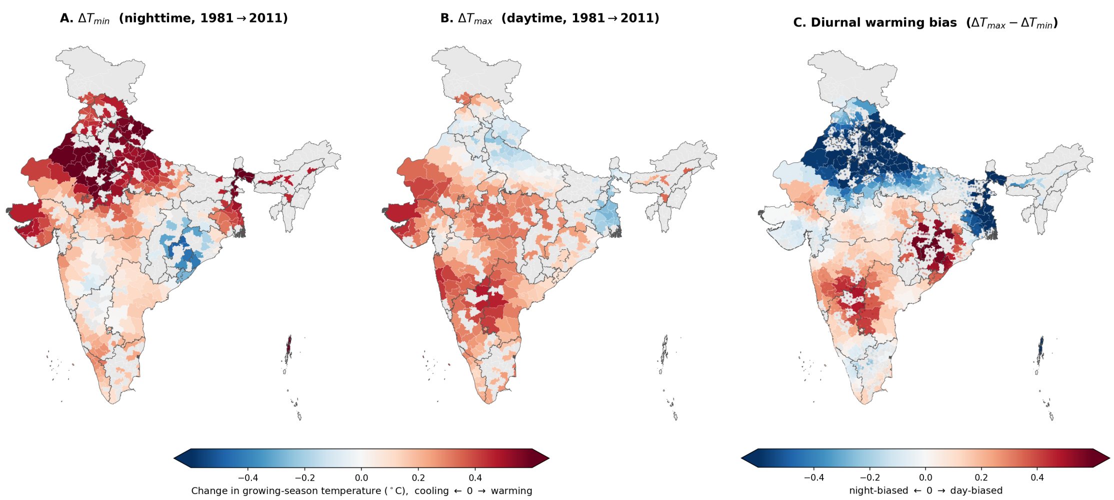
Notes: Reproduces the worker-share decomposition for urban households in the NSS Employment–Unemployment rounds (IPUMS-International, 1987–2009), classified to census-comparable categories: cultivators (self-cultivators and unpaid family workers), agricultural wage labour, and non-agriculture (everything else), the last split into informal (construction, mining, and low-skill services) and formal (manufacturing and high-skill services). Decadal panel with district and decade fixed effects and a state × decade linear trend, and precipitation controls; T_{min} and T_{max} are growing-season means over the preceding decade. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

FIGURE A1. Thermal Zones and Physiological Limits



Notes: Kernel densities of long-run surface-air minimum and maximum temperatures across the Census town sample (top) and the NSS/IPUMS urban household sample (bottom), with the night (20°C) and day (31°C) thresholds marked.

FIGURE A2. The diurnal warming geography: change in night, day, and the diurnal range



Notes: Long-difference change (1981–91 vs. 2001–11) in growing-season surface-air temperature across Census districts. Panel A is the change in the nighttime minimum (ΔT_{min}), Panel B the change in the daytime maximum (ΔT_{max}), and Panel C the diurnal warming bias ($\Delta T_{max} - \Delta T_{min}$): blue marks *night-biased* districts where the diurnal range is compressing (nights warming faster) and red *day-biased* districts where it is widening. Each panel uses a diverging scale centered at zero, so cooling reads blue and warming red; districts without Census+IMD coverage are grey. Census towns are overlaid on Panel C as translucent dots colored by their own bias. The night-biased belt is the Indo-Gangetic north and northwest; the day-biased belt is the peninsular Deccan and south.

Appendix D. Temperature Distributions and Maps

Appendix E. Monte Carlo Inference

To choose the panel estimator I ask which standard-error and trend specification has correct *size*: how often it rejects a true null. In each of 1,000 simulations I replace the climate regressor with a placebo built by randomly permuting the real climate series across districts—the T_{min}, T_{max}, and rainfall block is moved together, so the placebo keeps the regressor's decadal persistence and cross-variable correlation but is unrelated to the outcome by construction. Because the permutation reassigns whole series across units, it breaks the regressor's *spatial* autocorrelation; the exercise therefore calibrates size against persistence, not against spatial dependence, which I treat separately with the Conley standard errors reported throughout. I regress the real outcome on this placebo and record whether the joint null that both temperature coefficients are zero is rejected at the 5% level; under valid inference the rejection rate should be about 5%. Table A17 reports these rates for the Census-town outcomes, using the baseline surface-air growing-season regressor set; the Census-district and ICRISAT panels behave identically and are omitted for brevity. The pattern is clear: because the placebo is persistent, unadjusted and heteroskedasticity-robust errors over-reject by a factor of three or more, and a purely spatial Conley correction does not fix this; clustering by district brings the size close to nominal, and adding state-specific linear trends (or state-by-year fixed effects) keeps it near 5% while district-specific trends over-absorb. This motivates the preferred specification used throughout—district-clustered standard errors with state linear trends. Because the town panel nests towns within district-years, it additionally identifies the district-by-year fixed-effects row that saturates the district-level panel; it too holds size near nominal under the preferred estimator.

TABLE A17. Monte Carlo rejection rates: Census town outcomes

	Seasonal	Cultivators	Ag. labour	Non-ag.	Total workers
No adjustment (homoskedastic)	0.088	0.144	0.126	0.121	0.127
Heteroskedasticity-robust	0.073	0.106	0.098	0.093	0.080
Cluster by district	0.040	0.052	0.052	0.054	0.048
Cluster by district + district linear trends	0.045	0.041	0.044	0.058	0.036
Cluster by district + state linear trends	**0.037**	**0.055**	**0.041**	**0.059**	**0.048**
Cluster by district + state-by-year FE	0.039	0.048	0.047	0.064	0.048
Cluster by district + district-by-year FE	0.053	0.046	0.059	0.060	0.049
Cluster by state	0.083	0.093	0.068	0.077	0.073
Conley spatial HAC (50 km)	0.154	0.192	0.178	0.168	0.154
Conley spatial HAC (100 km)	0.150	0.209	0.182	0.180	0.158
Conley spatial HAC (200 km)	–	–	–	–	–
Conley spatial HAC (50 km, 2 lags)	0.074	0.106	0.106	0.099	0.110
Conley spatial HAC (50 km, 4 lags)	0.043	0.049	0.050	0.051	0.049
Replications			1000		

Notes: Each cell is the share of 1,000 simulations in which the joint null that both temperature coefficients equal zero is rejected at the 5% level, for the Census-town worker-share and log-employment outcomes. In each simulation the placebo regressor is built by randomly permuting the real climate series (the T_{min}/T_{max}/rainfall block, moved together) across towns, preserving its 10-year moving-average persistence and spatial distribution while breaking any true link to the outcome; a correctly sized estimator therefore rejects in roughly 5% of simulations. Rows vary the standard-error and trend specification, for the baseline surface-air growing-season regressor set; standard errors cluster at the district level. Conley rows sweep the spatial radius (50–200 km) at zero time lags and, at 50 km, the number of decadal time lags. Because towns nest within district-years, the district-by-year fixed-effects row is estimable here (unlike the district-level panel); the 200 km Conley row is omitted (–) as the spatial-HAC estimator over the full set of towns is computationally prohibitive. The preferred estimator—clustering by district with state-specific linear trends (bold)—holds size near 5%, whereas unadjusted and heteroskedasticity-robust standard errors over-reject. The Census-district and ICRISAT panels behave identically.